Ethan Havertape
CIS 344 Exam 2 Take Home

1.
H0 : mean income male = mean income female
H1: mean income male != mean income female

I ran a linear regression model to test if household income differs based on the gender of the head of the household. I had to first create dummy variables for Head as it was categorical. (Male = 1, Female = 0/ref group).

After running the test, I got a very large p-value of 0.802, anR^2 approximately at 0, and 95% confidence interval that included 0.

Therefore, we fail to reject the null hypothesis and conclude that head household gender does not have an impact on household income.

*SPSS output included below

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.006[a]	.000	-.001	27718.497

a. Predictors: (Constant), Head=Male
b. Dependent Variable: Income

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	48322547.012	1	48322547.012	.063	.802[b]
	Residual	1.270E+12	1653	768315067.23		
	Total	1.270E+12	1654			

a. Dependent Variable: Income
b. Predictors: (Constant), Head=Male

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	65167.174	1134.443		57.444	<.001	62942.078	67392.270		
	Head=Male	-355.831	1418.857	-.006	-.251	.802	-3138.778	2427.116	1.000	1.000

a. Dependent Variable: Income

2

I ran a bivariate correlation test to determine which variable is most appropriate to model WrkHrs. Additionally, I made a scatter/dot graph to visualize the correlation of all the variables to WrkHrs.

The highest correlation with WrkHrs was with Alcohol at 0.461. This indicates we have a moderately strong positive relationship where as alcohol spending increases, the number of hours worked per week also increases. The relationship is also statistically significant with a p-value <.001.

This is backed by the scatter plots where Alcohol and WrkHrs was the only relationship with any trend.

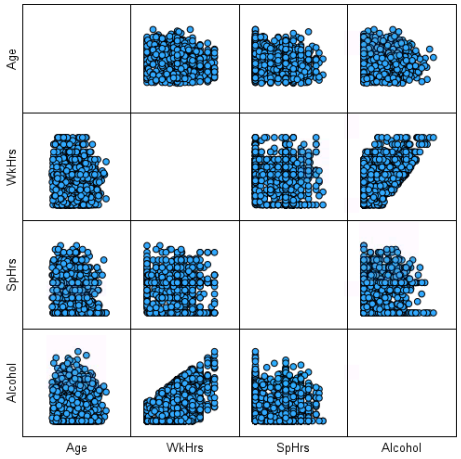
*Output included below

Correlations

		WkHrs	Age	SpHrs	Alcohol
WkHrs	Pearson Correlation	1	-.110***	-.153***	.461***
	Sig. (2-tailed)		<.001	<.001	<.001
	N	1655	1655	1655	1655
Age	Pearson Correlation	-.110***	1	-.105***	-.040
	Sig. (2-tailed)	<.001		<.001	.103
	N	1655	1655	1655	1655
SpHrs	Pearson Correlation	-.153***	-.105***	1	-.060*
	Sig. (2-tailed)	<.001	<.001		.014
	N	1655	1655	1655	1655
Alcohol	Pearson Correlation	.461***	-.040	-.060*	1
	Sig. (2-tailed)	<.001	.103	.014	
	N	1655	1655	1655	1655

***. Correlation at 0.001(2-tailed)

*. Correlation is significant at the 0.05 level (2-tailed).



3.

I set up this problem by creating a multiple linear regression model where my independent variables were Educ and WkHrs, and the dependent variable was income. The model was statistically significant with a p-value <.001 and a strong adjusted R^2 of 0.527.

The best-fit line equation is:

$$\text{Income} = -31{,}812.139 + 4{,}562.153(\text{Educ}) + 824.066(\text{WkHrs})$$

Both Educ and WkHrs are statistically significant with p-values <.001. For each additional level of education, income increases by about $4,562, holding WkHrs constant. For each additional hour worked, income increases by about $824, holding education constant.

12 Educ & 40 WkHrs = $55,896.34 income
16 Educ & 30 WkHrs = 65,904.29 income

This difference of ~$10,000 suggests a meaningful difference in average income. Education has a larger effect on income than hours worked.

*SPSS output included below

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.726[a]	.527	.527	19063.279

a. Predictors: (Constant), WkHrs, Educ
b. Dependent Variable: Income

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.697E+11	2	3.349E+11	921.445	<.001[b]
	Residual	6.004E+11	1652	363408602.54		
	Total	1.270E+12	1654			

a. Dependent Variable: Income
b. Predictors: (Constant), WkHrs, Educ

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-31812.139	2520.284		-12.622	<.001	-36755.427	-26868.852		
	Educ	4562.153	160.714	.480	28.387	<.001	4246.928	4877.377	1.000	1.000
	WkHrs	824.066	26.064	.535	31.617	<.001	772.945	875.188	1.000	1.000

a. Dependent Variable: Income

4.
I ran a multiple linear regression model to determine whether the model would be improved by including an interaction term. I did this by combining WkHrs and Educ by multiplying them together.

The model produced an adjusted R^2 of 0.532, compared to 0.527 (without the interaction term). This indicates that the interaction term produced a slight increase in explaining the variation.

EducWkHrs (interaction term) had a coefficient of 40.213 and was statistically significant since the p-value was less than 0.05.

Based on the General Linear F-test, the interaction term being statistically significant indicates that including it improves the model significantly.

This means that the effect of education on household income depends on the number of hours worked per week.

*SPSS output included below

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.730[a]	.533	.532	18948.977

a. Predictors: (Constant), EducWkHrs, Educ, WkHrs

b. Dependent Variable: Income

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.773E+11	3	2.258E+11	628.727	<.001[b]
	Residual	5.928E+11	1651	359063724.96		
	Total	1.270E+12	1654			

a. Dependent Variable: Income

b. Predictors: (Constant), EducWkHrs, Educ, WkHrs

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-8881.447	5597.010		-1.587	.113	-19859.434	2096.540		
	Educ	2930.031	390.421	.308	7.505	<.001	2164.259	3695.803	.167	5.975
	WkHrs	258.237	126.191	.168	2.046	.041	10.725	505.748	.042	23.735
	EducWkHrs	40.213	8.777	.416	4.582	<.001	22.997	57.429	.034	29.144

a. Dependent Variable: Income

5.

I ran a multiple linear regression model to first produce outputs including Educ, WkHrs, and Head of household Gender. In the model, gender, Head_Male, returned statistically significant with a p-value of 0.037. Therefore, gender has an effect on household income when we are controlling for amount of education and hours worked.

I then ran a similar multiple linear regression model but this time including sector of employment as dummy variables. After running the model including sector, the gender variable was no longer statistically significant (0.601).

This means that the initial effect of gender in the first output was due to differences in the sectors. Once I controlled for sectors, gender no longer had a significant effect on income. Therefore, income is best explained by just education and hours worked, rather than head gender and sector.

*Both SPSS outputs included below

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.727[a]	.529	.528	19043.982

a. Predictors: (Constant), Head=Male, Educ, WkHrs

b. Dependent Variable: Income

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.713E+11	3	2.238E+11	616.992	<.001[b]
	Residual	5.988E+11	1651	362673249.89		
	Total	1.270E+12	1654			

a. Dependent Variable: Income

b. Predictors: (Constant), Head=Male, Educ, WkHrs

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-30659.408	2577.688		-11.894	<.001	-35715.290	-25603.526		
	Educ	4566.345	160.564	.481	28.439	<.001	4251.415	4881.276	.999	1.001
	WkHrs	826.350	26.060	.536	31.709	<.001	775.236	877.465	.998	1.002
	Head=Male	-2035.040	975.774	-.035	-2.086	.037	-3948.926	-121.155	.998	1.002

a. Dependent Variable: Income

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.749[a]	.561	.560	18378.813

a. Predictors: (Constant), Sector=3.0, Educ, WkHrs, Head=Male, Sector=2.0

b. Dependent Variable: Income

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	7.131E+11	5	1.426E+11	422.210	<.001[b]
	Residual	5.570E+11	1649	337780753.76		
	Total	1.270E+12	1654			

a. Dependent Variable: Income

b. Predictors: (Constant), Sector=3.0, Educ, WkHrs, Head=Male, Sector=2.0

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-30071.789	2489.397		-12.080	<.001	-34954.501	-25189.076		
	Educ	4673.946	155.737	.492	30.012	<.001	4368.483	4979.409	.989	1.011
	WkHrs	867.891	25.473	.563	34.071	<.001	817.928	917.854	.973	1.028
	Head=Male	507.151	970.607	.009	.523	.601	-1396.600	2410.902	.939	1.064
	Sector=2.0	-9224.161	990.671	-.164	-9.311	<.001	-11167.267	-7281.055	.855	1.169
	Sector=3.0	-13709.634	1556.219	-.155	-8.810	<.001	-16762.007	-10657.260	.861	1.161

a. Dependent Variable: Income